

Step 1: Create Database

```
DROP DATABASE IF EXISTS student_db3;
```

```
CREATE DATABASE student_db3;
```

```
USE student_db3;
```

OUTPUT:-

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	11:13:43	DROP DATABASE IF EXISTS student_db3	1 row(s) affected	0.032 sec
2	11:13:46	CREATE DATABASE student_db3	1 row(s) affected	0.047 sec
3	11:13:48	USE student_db3	0 row(s) affected	0.000 sec

STEP 2:- Create Table

```
DROP TABLE IF EXISTS student_marks11;
```

```
CREATE TABLE student_marks11 (  
    roll_no INT PRIMARY KEY,  
    name VARCHAR(50),  
    subject VARCHAR(50),  
    marks DECIMAL(5,2)  
);
```

OUTPUT:-

Action Output				
#	Time	Action	Message	Duration / Fetch
1	11:25:49	DROP TABLE IF EXISTS student_marks11	0 row(s) affected	0.031 sec
2	11:25:52	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	0 row(s) affected	0.031 sec
3	11:25:54	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	Error Code: 1050. Table 'student_marks11' already exists	0.000 sec
4	11:25:56	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	Error Code: 1050. Table 'student_marks11' already exists	0.000 sec

Step 3: Insert Student Records

```

INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES
(1, 'Ravi', 'Math', 85.50),
(2, 'Sita', 'Math', 92.75),
(3, 'Anil', 'Math', 78.40),
(4, 'Priya', 'Math', 88.90),
(5, 'Vijay', 'Math', 80.25),
(6, 'Subbusir', 'AWS', 98.50),
(7, 'Dwaraka', 'DBMS', 95.75),
(8, 'Ranjani', 'English', 97.40),
(9, 'Kavitha', 'AWS', 99.90),
(10, 'Seetha', 'Azure', 82.25);

```

OUTPUT:-

```

9
10 DROP TABLE IF EXISTS student_marks11;
11
12 CREATE TABLE student_marks11 (
13     roll_no INT PRIMARY KEY,
14     name VARCHAR(50),
15     subject VARCHAR(50),
16     marks DECIMAL(5,2),
17 );
18
19 INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES
20 (1, 'Ravi', 'Math', 85.50),
21 (2, 'Sita', 'Math', 92.75),
22 (3, 'Anil', 'Math', 78.40),
23 (4, 'Priya', 'Math', 88.90),
24 (5, 'Vijay', 'Math', 80.25),
25 (6, 'Subbusir', 'AWS', 98.50),
26 (7, 'Dwaraka', 'DBMS', 95.75),
27 (8, 'Ranjani', 'English', 97.40),
28 (9, 'Kavitha', 'AWS', 99.90),
29 (10, 'Seetha', 'Azure', 82.25);
30
31 SELECT * FROM student_marks11;
32 SELECT COUNT(*) AS total_students FROM student_marks11;

```

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	11:55:21	DROP TABLE IF EXISTS student_marks11	0 row(s) affected	0.016 sec
✓ 2	11:55:21	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VAR...	0 row(s) affected	0.031 sec
✓ 3	11:55:21	INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES (1,'Ravi','Math',85.50), (2,...	10 row(s) affected	0.047 sec
✓ 4	11:55:21	SELECT * FROM student_marks11	10 row(s) returned	0.000 sec / 0.000 sec
✓ 5	11:55:21	SELECT COUNT(*) AS total_students FROM student_marks11	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	Subbusir	AWS	98.50
7	Dwaraka	DBMS	95.75
8	Ranjani	English	97.40
9	Kavitha	AWS	99.90
10	Seetha	Azure	82.25

total_students

10

Step 4: Display All Records

27

28 • `SELECT * FROM student_marks11;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	3	Anil	Math	90.00
	4	Priya	Math	88.90
	6	Subbusir	AWS	98.50
	7	Dwaraka	DBMS	95.75
	8	Ranjani	English	97.40
	9	Kavitha	AWS	99.90
	10	Seetha	Azure	82.25

student_marks11 27 x

Read Only Context Help Snippets

OUTPUT:-

student_marks11 27 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:42:44	SELECT * FROM student_marks11 LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

Step 5: Count Students

```
SELECT COUNT(*) AS total_students
FROM student_marks11;
```

OUTPUT:-

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:44:51	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
2	11:44:55	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Step 6: Update Records

```

UPDATE student_marks11
SET marks = 90.00
WHERE roll_no = 3;

UPDATE student_marks11
SET subject = 'Remedial Math'
WHERE marks < 80;

```

OUTPUT:-

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	11:48:25	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
2	11:48:29	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.015 sec
3	11:48:30	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
4	11:48:32	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec

Step 7: Delete Records

```

SELECT * FROM student_marks11;

DELETE FROM student_marks11
WHERE roll_no = 5;

DELETE FROM student_marks11
WHERE marks < 75;

```

OUTPUT:-

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	11:51:56	DELETE FROM student_marks11 WHERE roll_no = 5	0 row(s) affected	0.000 sec
2	11:51:58	DELETE FROM student_marks11 WHERE roll_no = 5	0 row(s) affected	0.000 sec
3	11:52:01	DELETE FROM student_marks11 WHERE marks < 75	0 row(s) affected	0.000 sec
4	11:52:03	DELETE FROM student_marks11 WHERE marks < 75	0 row(s) affected	0.000 sec

Step 8: Sort Records

```

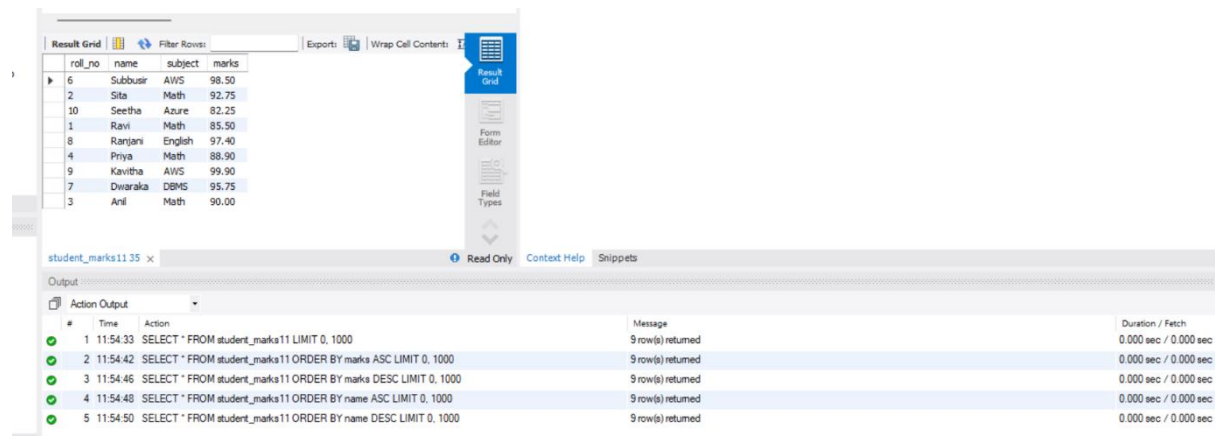
SELECT * FROM student_marks11;

SELECT * FROM student_marks11 ORDER BY marks ASC;
SELECT * FROM student_marks11 ORDER BY marks DESC;
SELECT * FROM student_marks11 ORDER BY name ASC;
SELECT * FROM student_marks11 ORDER BY name DESC;

SELECT subject, SUM(marks) AS total_marks
FROM student_marks11
GROUP BY subject;

```

OUTPUT:-



The screenshot shows a database management tool interface. At the top, a 'Result Grid' displays a table with columns 'roll_no', 'name', 'subject', and 'marks'. The table contains 10 rows of student data. Below the table, there is a tab labeled 'student_marks11 35'. The main area shows an 'Output' window with a table of execution logs. The logs show five SQL queries being executed, each returning 9 rows. The queries are: 1. SELECT * FROM student_marks11 LIMIT 0, 1000; 2. SELECT * FROM student_marks11 ORDER BY marks ASC LIMIT 0, 1000; 3. SELECT * FROM student_marks11 ORDER BY marks DESC LIMIT 0, 1000; 4. SELECT * FROM student_marks11 ORDER BY name ASC LIMIT 0, 1000; 5. SELECT * FROM student_marks11 ORDER BY name DESC LIMIT 0, 1000.

roll_no	name	subject	marks
6	Subbusir	AWS	98.50
2	Sita	Math	92.75
10	Seetha	Azure	82.25
1	Ravi	Math	85.50
8	Ranjani	English	97.40
4	Priya	Math	88.90
9	Kavitha	AWS	99.90
7	Dwaraka	DBMS	95.75
3	Anil	Math	90.00

#	Time	Action	Message	Duration / Fetch
1	11:54:33	SELECT * FROM student_marks11 LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
2	11:54:42	SELECT * FROM student_marks11 ORDER BY marks ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
3	11:54:46	SELECT * FROM student_marks11 ORDER BY marks DESC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
4	11:54:48	SELECT * FROM student_marks11 ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
5	11:54:50	SELECT * FROM student_marks11 ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

Step 9: Aggregate Functions

```

SELECT subject, AVG(marks) AS avg_marks
FROM student_marks11
GROUP BY subject;

```

```

SELECT subject, COUNT(*) AS num_students
FROM student_marks11
GROUP BY subject;

```

OUTPUT:-

Result 40 x					Read Only	Context Help	Snippets
Output							
Action Output							
#	Time	Action	Message	Duration / Fetch			
1	11:56:42	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec			
2	11:56:45	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec			
3	11:56:48	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec			
4	11:56:53	SELECT subject, COUNT(*) AS num_students FROM student_marks11 GROUP BY subject LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec			

Step 10: HAVING Clause

```
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks11
GROUP BY subject
HAVING AVG(marks) > 90;
```

```
SELECT subject, COUNT(*) AS num_students
FROM student_marks11
GROUP BY subject
HAVING COUNT(*) > 1;
```

OUTPUT:-

Result 47 x					Read Only	Context Help	Snippets
Output							
Action Output							
#	Time	Action	Message	Duration / Fetch			
2	11:58:11	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject HAVING AVG(marks) > 90	3 row(s) returned	0.000 sec / 0.000 sec			
3	11:58:13	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject HAVING AVG(marks) > 90	3 row(s) returned	0.000 sec / 0.000 sec			
4	11:58:17	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject HAVING AVG(marks) > 90	3 row(s) returned	0.016 sec / 0.000 sec			
5	11:58:21	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject HAVING AVG(marks) > 90	3 row(s) returned	0.000 sec / 0.000 sec			
6	11:58:23	SELECT subject, AVG(marks) AS avg_marks FROM student_marks11 GROUP BY subject HAVING AVG(marks) > 90	3 row(s) returned	0.000 sec / 0.000 sec			
7	11:58:29	SELECT subject, COUNT(*) AS num_students FROM student_marks11 GROUP BY subject HAVING COUNT(*) > 1	2 row(s) returned	0.000 sec / 0.000 sec			